MOCK B

Advanced Financial Management

September 2024

AFM

Time allowed: **3 hours 15 minutes**

This question paper is divided into **two** sections:

Section A: This **ONE** question is compulsory and **MUST** be attempted

Section B: **BOTH** questions are compulsory and **MUST** be attempted

Formulae and tables are on pages 10 to 13.

Do NOT open this paper until instructed by the supervisor

This question paper must not be removed from the examination hall

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SECTION A

THIS ONE QUESTION IS COMPULSORY AND MUST BE ATTEMPTED

1 The following exhibits, available on the left-hand side of the screen (in the CBE exam), provide information relevant to the question.

1 **Marvin Co expansion plans**

2 **Financial details of the project**

3 **Follow-on option**

This information should be used to answer the question requirements within your chosen response option(s).

Exhibit 1: Marvin Co expansion plans

Marvin Co is a manufacturer of vacuum cleaners. It is based in the United States of America where the functional currency is the US dollar ($).

The directors have decided to expand into South America, by undertaking a five-year project investing in a new manufacturing facility in Lanavia, where the currency is the Lanavia peso (LP).

Lanavia's economy has grown very quickly over the last decade as government incentives have encouraged inward investment, particularly in the manufacturing and technology sectors. Consequently, there are a large number of highly skilled workers in Lanavia, and many modern ('state-of-the-art') manufacturing facilities.

Initial market research carried out over the last few months costing $500,000 has revealed that there is a relatively small market for vacuum cleaners in Lanavia at the moment, but that the market is expected to grow rapidly over the next few years.

Exhibit 2: Financial details of the project

Marvin Co intends to invest 3 million LP to buy a factory in Lanavia, and then another 1 million LP to buy the state-of-the-art manufacturing machines and equipment. These amounts will be paid immediately.

After five years, the machines and equipment will have a realisable value of 50,000 LP, and the factory will have a realisable value of 2.8 million LP (both in money terms).

Working capital of 0.5 million LP will also be required immediately. This will be increased by the rate of inflation in Lanavia throughout the project, and will then be released in the final year of the project.

Sales in Lanavia are forecast to be:

	Year 1	Year 2	Year 3	Year 4	Year 5
Sales (units)	15,000	26,000	32,000	35,000	33,000

In year 1, the unit selling price for the vacuum cleaners will be 120 LP, and the directors intend to increase the price by 5% each year thereafter. The unit variable manufacturing cost in year 1 is expected to be 45 LP, and the total fixed cost in year 1 is expected to be 1.5 million LP. These costs are expected to inflate in line with inflation in Lanavia.

Also, net contribution in the USA is expected to increase by $0.12 million before tax in year 1 if the Lanavia project goes ahead, as the production process in the USA is modified to incorporate some of the new state-of-the-art Lanavian manufacturing methods. This annual amount is expected to increase over the life of the project in line with the inflation rate in the USA.

Corporation tax

The tax rate in Lanavia is 15%, and the tax rate in the USA is 25%. In both countries, tax is payable in the year the tax liability arises. Both countries' tax authorities allow losses to be carried forward and written off against future profits for taxation purposes. A bi-lateral tax treaty exists between Lanavia and the USA.

Tax allowable depreciation will be available on the manufacturing machines and equipment at the rate of 20% on original cost per year. In the year of disposal, a balancing allowance or charge will be claimed. The factory will not be eligible for tax allowable depreciation.

Other information

The spot exchange rate is 2.58 LP to $1.

Inflation is expected to be 3% per year in the USA and 6% per year in Lanavia for the foreseeable future.

The appropriate cost of capital for a project with this level of risk is 10%.

Exhibit 3: Follow-on option

The initial Lanavia investment is only a relatively short-term, five-year project. However, the directors expect that if the project is successful, it will allow Marvin Co to exploit a follow-on opportunity in Lanavia by investing in a second factory.

Initial investigations into this follow-on project suggest that an investment of $2 million in four years' time would yield an overall after-tax return of $0.3 million each year in perpetuity from year 5 onwards. The standard deviation of the follow-on project's cash flows is likely to be 30% and the risk free rate of return is currently 6%. The business risk associated with the follow-on project is expected to be very similar to the business risk associated with the initial project.

Required:

(a) Prepare a report for the Board of Directors of Marvin Co in which you:

(i) Calculate the net present value of the proposed initial investment in Lanavia, without considering the follow-on option to invest in the second factory. Show all relevant calculations. (17 marks)

(ii) Calculate the overall value of the project after taking into account the value of the follow-on option. (5 marks)

(iii) Discuss the approach taken and the assumptions made in parts (i) and (ii) above, and recommend whether or not Marvin Co should undertake the investment in Lanavia. (10 marks)

(b) The chief executive of Marvin Co recently attended a lecture on 'behavioural finance', where he was interested to discover that financial decisions made by companies are not always rational. He was particularly concerned because he realised that he and his fellow directors had shown signs of 'gambler's fallacy', 'anchoring', 'herd instinct' and 'confirmation bias' when making previous financial decisions.

Required:

Explain the meaning of these four behavioural finance terms. (8 marks)

Professional marks will be awarded for the demonstration of skill in communication, analysis and evaluation, scepticism and commercial acumen in your answer. (10 marks)

(Total: 50 marks)

SECTION B

BOTH QUESTIONS ARE COMPULSORY AND MUST BE ATTEMPTED

2 The following exhibits, available on the left-hand side of the screen (in the CBE exam), provide information relevant to the question.

1 **Grindall Co**

2 **Planned divestment**

3 **Financial information**

This information should be used to answer the question requirements within your chosen response option(s).

Exhibit 1: Grindall Co

Grindall Co is a large listed retail company based in the country of Haim. It built its reputation as a food retailer, but in recent years it has diversified into furniture. It now owns several furniture stores throughout the country. These furniture stores are forecast to generate a profit after tax of $35.3 million in the coming financial year ('Year 1'), which will be 25% of the overall company total profit after tax.

Exhibit 2: Planned divestment

Over the last year or so, a deep recession has hit Haim, and the directors are concerned that the furniture stores will suffer. Therefore they are considering selling the furniture business off quickly. In order to achieve a quick sale, the finance director estimates that a premium of only 10% over the current free cash flow valuation of the furniture business will be achievable.

The other directors are keen to explore the possibility of selling off the furniture business quickly, so they have asked the finance director to calculate the likely selling price of the furniture business, and to consider the likely impact of the sale on:

- Grindall Co's statement of financial position,
- its earnings per share in Year 1, and
- its weighted average cost of capital.

If the sale can be completed quickly, the finance director's plan is to use the sale proceeds to strengthen the financial position of the remaining business by:

1 repaying Grindall Co's 5% coupon bonds,

2 keeping $50 million in cash, and

3 investing the remainder of the sale proceeds in non-current assets.

The current tax rate on profits is 25%. The risk-free rate is 3% and the market risk premium is 8%.

Exhibit 3: Financial information

Valuation of the furniture business

The current free cash flow valuation of the furniture business, using the current weighted average cost of capital of 10%, and based on an assumption that cash flows will grow at 1% per year in perpetuity, is $347.6 million.

Grindall Co's current summarised statement of financial position

Assets	$m
Non-current assets	692.4
Current assets	329.1
Total assets	1,021.5
Equity and liabilities	
Share capital	400.0
Retained earnings	210.9
Total equity	610.9
Non-current liabilities	
5% bonds	200.0
Bank loans	130.0
Total non-current liabilities	330.0
Current liabilities	80.6
Total liabilities	410.6
Total equity and liabilities	1,021.5

Other relevant information

- The 5% bonds are trading at par.
- The current net book value of the non-current assets of the furniture business to be sold can be assumed to be $230 million. The profit on the sale of the furniture business should be taken directly to reserves.
- Additional investment in current assets is expected to earn a 3% pre-tax return and additional investment in non-current assets is expected to earn an 15% pre-tax return.
- The average asset beta for listed food retailing companies in Haim is 0.8. Assume that the debt beta is zero.
- Grindall Co currently has 400 million $1 shares in issue. These are currently trading at $1.60 per share. The finance director expects the share price to rise by 5% once the sale has been completed, as she thinks that the stock market will perceive it to be a good deal.
- The interest rate on the bank loans is 7%.

Required:

(a) Calculate the expected sales price of the furniture business and demonstrate its impact on Grindall Co's statement of financial position, forecast earnings per share and weighted average cost of capital. (13 marks)

(b) Evaluate the decision to sell the furniture business. (7 marks)

Professional marks will be awarded for the demonstration of skill in analysis and evaluation, scepticism and commercial acumen in your answer. (5 marks)

(Total: 25 marks)

3 The following exhibits, available on the left-hand side of the screen (in the CBE exam), provide information relevant to the question.

1 **Economic information**

2 **Rachida SA – two transactions**

3 **Hedging information**

This information should be used to answer the question requirements within your chosen response option(s).

Exhibit 1: Economic information

Over the last decade or so, the major global economies have been in turmoil. Interest rates were initially slashed by central banks all over the world in an effort to minimise the impact of the global financial crisis. This in turn affected many currency exchange rates around the world. Supply and demand for individual currencies fluctuated dramatically as investors sought safe currencies to invest in. However, in recent times, rising oil prices and food prices have led to an increase in inflation, so most central banks have started raising interest rates again to keep inflation under control. Both interest rates and currency rates are likely to become more volatile in the short to medium term for many major trading countries.

Exhibit 2: Rachida SA – two transactions

In this environment, the Board of Directors of Rachida SA, a French manufacturing company, is reviewing the company's interest rate and currency risk strategies for the next few months. The directors have identified two major transactions that they are considering hedging:

- Recent cash flow forecasts have identified the need to borrow €3,250,000 for a period of six months commencing in six months' time.
- Also, a major new piece of equipment is to be purchased from a supplier in Beeland (currency the B$), and Rachida SA will need to make a payment of B$2.15 million in 3 months' time.

Exhibit 3: Hedging information

Assume that it is now 1 December. Futures and options contracts may be assumed to expire at the end of the relevant month, and the company may be assumed to be able to borrow at the base rate of interest plus 1.00%. The base rate is currently 3.5%.

3 month futures prices (€500,000, 3 month contracts)

March	96.56
June	96.29

3 month options on futures prices (€500,000, 3 month contracts). Premiums are annual % amounts. Tick value is €12.50.

	CALLS		PUTS	
	March	June	March	June
96.25	0.445	0.545	0.085	0.185
96.50	0.280	0.390	0.170	0.280
96.75	0.165	0.265	0.305	0.405

Foreign exchange rates:

Spot	B$ 1.4692 – 1.4735 to €1
3 month forward	B$ 1.4632 – 1.4668 to €1

Currency option prices

Stock Exchange B$/€ options, contract size €31,250, premiums are in Beeland cents per €1:

	CALLS		PUTS	
Strike price (€1 = B$...)	March	June	March	June
1.4500	3.12	3.66	1.56	1.99
1.4600	2.55	2.95	1.99	2.51
1.4700	2.14	2.48	2.51	3.03

Required:

(a) Using the above information, illustrate the possible results of hedges with interest rate futures contracts, and with options over interest rate futures, if interest rates in six months' time increase by 0.75%. Recommend which hedge should be selected. **(12 marks)**

(b) Illustrate and discuss the possible outcomes of forward market and currency options hedges if the exchange rate in three months' time are B$1.4350 – B$1.4386 to €1. **(8 marks)**

Professional marks will be awarded for the demonstration of skill in analysis and evaluation, and commercial acumen in your answer. **(5 marks)**

(Total: 25 marks)

FORMULAE SHEET

Modigliani and Miller proposition 2 (with tax)

$$k_e = k^i_e + (1 - T)(k^i_e - k_d)\ \frac{V_d}{V_e}$$

Or rearranged

$$k_e + (1\text{-}T)\ k_d\ \frac{V_d}{V_e} = k^i_e + (1 - T)\ k^i_e\ \frac{V_d}{V_e}$$

The capital asset pricing model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{V_e + V_d(1-T))}\ \beta_e\right] + \left[\frac{V_d(1-T)}{V_e + V_d(1-T))}\ \beta_d\right]$$

The growth model

$$P_o = \frac{D_o(1+g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d}\right]k_e + \left[\frac{V_d}{V_e + V_d}\right]k_d(1-T)$$

The Fisher formula

$$(1+i) = (1+r)(1+h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_o \times \frac{(1+h_c)}{(1+h_b)} \qquad F_0 = S_0 \times \frac{(1+i_c)}{(1+i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_1}\right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_aN(d_1) - P_eN(d_2)e^{-rt}$$

MATHEMATICAL TABLES

Present value table

Present value of 1, i.e. $(1 + r)^{-n}$

where r = discount rate

n = number of periods until payment

Periods (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239

Periods (n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.206	0.194
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.933
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065

Annuity table

Present value of an annuity of 1, i.e. $\frac{1-(1+r)^{-n}}{r}$

where r = interest rate

n = number of periods

Periods (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	.1783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.893	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606

Periods (n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.496	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.586	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675